

Diagnosis of Middle Ear Diseases using Deep Learning Paradigm

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Abstract— In recent years, research has focused on developing a deep learning network that could use images of the ear drum identify conditions in the middle ear. Automatic ear problem diagnosis in particular, can be helpful. Even when antibiotics are used to treat it, otitis media still drives hearing impairment, even loss of hearing in almost all age groups. The evaluation of the tympanic membrane and assessment of the potential value of the network during the diagnostic process constitute the initial examination for the diagnosis of ear sickness. The strategy for identifying middle ear disorders using several deep learning models is proposed in this paper. Deep neural learning aid in the analysis of ear condition may increase medical accessibility for persons without access to otolaryngologists by assisting non-specialists in recognizing otitis media. This deep learning network will be able to diagnose the middle ear problems more precisely and help doctors analyse images of the tympanic membrane.

Keywords- Deep learning, Convolutional Neural Network, Machine learning, Otitis media, Artificial Neural Network.

I. INTRODUCTION

A common ailment that can be easily cured with quick medical attention is ear and mastoid disease. However, if the illness is not identified and properly treated, it may result in long-term effects including hearing loss. The assessment of ear condition usually commences with a substantial examination using traditional otoendoscopy or otoscopy, in addition to a description of the patient's condition [4]. A number of medical specialties have adopted and utilised AI technologies. Medical practise has been adversely affected by recent advancements in image interpretation and diagnosis. Recent studies have shown favorable performance in the analysis of a variety of medical pictures, including computed tomography scans, mammography, and chest X-rays. It can be very useful to use automatic diagnostics for ear conditions. Despite the use of antibiotics, otitis media causes hearing loss

and is a significant health burden, especially in children and the elderly[3]. The initial step for the diagnosis of ear condition is the analysis of the eardrum. The inspection of the eardrum is the initial trail in the diagnosis of ear illness. However, in the absence of an attending physician, a circumstance that commonly occur in treatment centers and remote clinics, it might be challenging to carry out this evaluation accurately. The disease of the middle ear and external ear (also known as pinna) is the persistent cause for giving antibiotic prescriptions and performing surgery on children and young people. Most of the time, imaging examinations is not necessary; instead, they are only ordered when problems are suspected or a treatment is not working. By helping nonspecialists to diagnose otitis media, deep learning networks may increase access to healthcare for those who do not have access to otolaryngologists. Patients won't benefit from extra diagnostic imaging studies in the great majority of these cases because a correct diagnosis can typically be obtained only by clinical evaluation[2,3]. As a result, tomography is crucial, especially in illnesses that are complex and recur; the results of imaging may have a big influence on how to proceed with treatment [1]. The goal of the work is to create a diagnostic system that can more accurately identify otitis media (OM) utilising images of the eardrum (tympanic membrane) and machine learning to automatically extract specific features and perform image classification. If OM is present, this diagnostic system will give a survey volunteer trustworthy information to suggest the patient or his family to see an ENT or seek expert aid.

II. LITERATURE REVIEW

Dongchu et.al.[19], Around 9 public CNN were trained to categorize the auditory canal and eardrum into 6 types of ear diseases. An ensembler classifier was created by incorporating the scores of the two models whose performance was outstanding. Transfer learning models were selected based on

the training time and accuracy, and using those two models ensemble classifiers gave a compelling increase in the models, with an efficiency of 93-67%. This study demonstrates the value of applying a deep neural network to the clinical setting for the diagnosis and management for disorders in the ear.

Agnieszka et.al.[20], The preferred technique is high-resolution CT (HRCT) of the temporal bone. Imaging investigations were employed since the atretic external ear prevents viewing of eardrum, middle-ear components. Understanding the entire diagnosis and radiological cues helps to restrict the clinical assessment and reduces errors. The doctor and radiologist must work closely together to assess temporal bone diseases.

Michelle et.al.[21], The algorithm is given a RGB image, which predicts whether of the four ear classes chronic OM, earwax plug, myringosclerosis, normal is existent. The model focuses on the specific area on the image that raises medical suspicion when the prediction is an ear condition. They trained a number of classification models utilizing the RGB image's extracted features in order to produce the predictions. The system attained a 93.9% accuracy rate, an 87.8% sensitivity rate, a 95.9% specificity rate, and an 87.7% positive predictive value.

Moreno et.al.[22], In this study, the performance of a few of the classifiers in cutting-edge machine learning and related fields after choosing a sizable number of datasets from the primary public repositories. However, the proposal's scope extended beyond determining the Responsiveness, Clarity, and Adaptive Precision values and included a statistical analysis supported by Holm statistical tests and Friedman.

Geert et.al.[23], The training process is substantially streamlined when the most common models are trained from beginning to end in a supervised manner. The technical skill in performing tympanocentesis on synthetic tympanic membranes was evaluated using an infant mannequin model. Simulator-based interactive continuing medical education programmes can improve knowledge, diagnostic precision, and therapy paradigms.

Michael et.al.[24], It was determined how to diagnose Acute OM, Glue ear, and a healthy tympanic membrane. The technical skill in performing tympanocentesis on synthetic larynx membranes was evaluated using an infant dummy model. The likelihood of AOM and OME being misdiagnosed frequently is raised by the evaluation of indicative ability with the video-conferred exam. Simulator-based interactive continuing medical education programmes may improve clinical judgement, treatment paradigms, and skill levels.

Zebin et.al.[25], Using two most commonly used deep learning models namely MobileNet-V2 and Xception otoscopic image classifier for pediatric otitis media. Both the models have similar accuracy of 97.4% and 95.7% and the accuracy of the models with the images captured by the smartphone were 90.6% and 88.5%.

Adi et.al.[26], In this paper, a unique computer aided decision model was developed based on Deep neural network. The consolidation of Channel and spatial model, hypercolumn approach, residual block were encapsulated

in the model in order to boost the ability of the model. It guaranteed the comprehensive accuracy of 98.26% along with 97.68% of sensitivity and 99.3% specificity. Comparing the suggested model with trained CNNs like VGG-Net, GoogLeNet, the outcomes were apparent.

Moreno et.al.[27], The authors of this paper calculated the effectiveness of compatible classifiers of machine learning on preferred count of datasets. They performed descriptive statistics along with calculating accuracy, sensitivity and specificity. For future work, they were planning to use classification algorithms on existing datasets to obtain a dataset of those diseases in which the hospitals are interested to test the effectiveness.

Yan-Mei et al. [28], This study aimed to create a deep neural network for identifying Chronic OM using computed tomography scans of the temporal bone. The domain of concern from 2-D CT scans was automatically retrieved using a deep learning system, and COM was diagnosed using this approach. This model's performance was on par with—and occasionally even better than—that of clinical experts. These findings suggested a bright future for the use of artificial intelligence in clinical settings to diagnose COM using CT images.

Ri Su et.al. [29], To increase the veracity of the classification and diagnosis of chronic middle ear diseases in clinical practise and decrease the incidence of missed diagnoses and misdiagnoses, the study aims to generate a model for automatic information extraction, classification, interpretation of chronic middle ear diseases. The temporal bone HRCT image data of patients can be automatically broken down into 3D data blocks containing middle ear features by the deep learning network model proposed in this study, which can reduce the overall size of the data while maintaining the relationship between corresponding images. 3D-CNN can then be used for CSOM and MEC classification and diagnosis.

Zeng et al. [30], In this paper, study was conducted on the middle ear characteristics by combining deep neural networks and a logistic regression model characteristics, to predict CHL from otoscopic pictures. The methodology used in this analytic investigation was more accurate than the logistic regression model and otologists in predicting CHL in ears with OME. This suggests that, in situations where CHL cannot be evaluated, there is significant potential for the application of artificial intelligence techniques to facilitate CHL evaluation.

III. METHODOLOGY

A. Dataset description

The dataset was acquired from the Department of ENT, AIIMS. The acquisition of imaging data for healthy and sick tympanic membranes was one of the main challenges encountered throughout the study. The dataset comprises approximately 250 photographs classified into 6 classes. The number of acute samples was 26, normal ear sample was counted 90, whereas glue ear, otomycosis and safe CSOM were 16, 21, 56 respectively. And remaining sample images

which did not fall under there 5 classes were categorized as others, which is the sixth class in the dataset.

Sr. No	Class	No. of images
1.	Acute Otitis Media (AOM)	26
2.	Normal Ear	90
3.	Glue Ear (Otitis Media Effusion- OME)	16
4.	Otomycosis	21
5.	Safe Chronic suppurative otitis media (CSOM)	56
6.	Others	29

TABLE-1 Dataset description

Acute Otitis Media (Fig. 1(A)), sometimes known as an ear infection, is the second-most frequent pediatric emergency room diagnosis after upper respiratory infections. Acute otitis media can happen at any age, but between 6 and 24 months is when it is most frequently diagnosed.[5]

In children, glue ear (Fig. 1(B)) is a prevalent cause of hearing loss. It can result in a hearing loss to a degree which can turn normal speech into a whisper (0 to 50 dB HL). Glue ear can be categorised as bilateral glue ear and unilateral glue ear. Bilateral glue ear can cause serious hearing deficit, which can lead to development issues in children, however unilateral glue ear does not cause serious problems.[6]

Chronic suppurative otitis media (Fig. 1(C)) (CSOM) is a condition marked by chronic mucoid otorrhoea, persistent tympanic membrane perforation, and serious inflammation cleft near middle ear . It's one of the significant reasons for avertible hearing deficit in the world. Most often, it is a result of inadequate treatment for AOM, which lowers one's quality of life.[7]

Otomycosis((Fig. 1(D)) is a slight mycotic septicity of the outer ear canal that otolaryngologists regularly encounter. It is typically detectable by a clinical examination. However, a strong index of suspicion is necessary for the proper diagnosis[8].

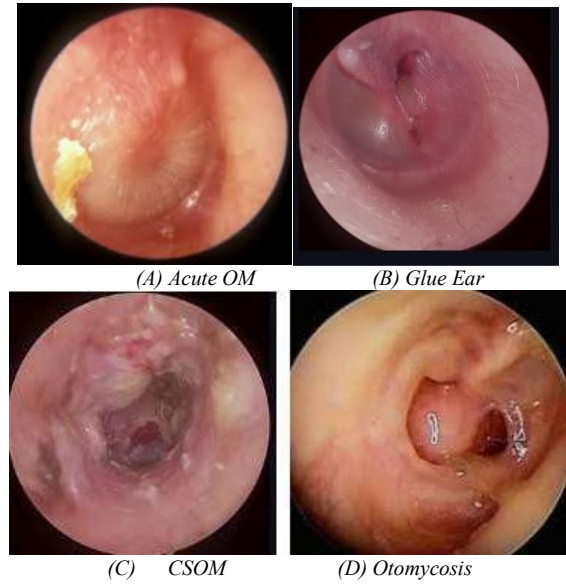


Fig.1 Dataset description

B. Convolutional Neural Network

CNN architecture is the most prevalent type of ANN architecture. CNNs are typically employed to tackle challenging image-driven pattern recognition problems and, thanks to their accurate yet straightforward architecture, provide a streamlined way to get started with ANNs. In this article, CNNs are briefly introduced, along with recently released papers and innovative methods for creating these amazingly outstanding image recognition models. The introduction makes the assumption that you are already familiar with the basics of ANNs and machine learning[15]. Convolution, pooling, dropout, and fully connected layers are the four successive layers that make up the deep learning framework CNN. Convolutional layer filters the image which is given as input and develops a featuremap with input picture (h) and filter (f).[3]

$$hXf = \sum \sum h(m, n)(f(x - m, y - n)) \quad (1)$$

The property map is increased by the convolutional layer of the CNN, which has multiple layers, which raises the calculation cost. A convolution layer with a stripe value t as 1, and size of filter d*d fiis applied to a size input image I of size N*N.

$$O = (N - t)/V + 1 \quad (2)$$

The output layer's size is determined by the equation. To lower the cost of computation, the convolutional layer is used before the pooling layer. By employing average or maximum functions to compress the result, the subregions are summarized. A fully connected layer is followed by numerous convolutional and pooling layers [fig.2] The processed extracted property vector generates as many kinds of output as input[9].

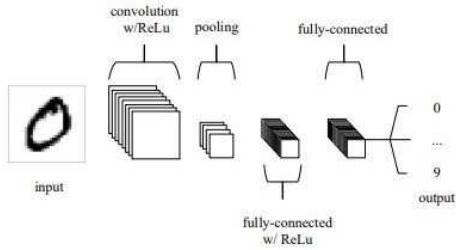


Fig.2 Architecture of CNN [35]

a. GoogLeNet

GoogLeNet is a 22 layers deep neural network model. Out of these 22 layers, 1 of them is a fully connected layer and the remaining 21 layers are convolutional layers. The model consists of three distinct filters of dimensions 5x5, 3x3, 1x1. An Inception module in GoogleNet features a pooling layer in addition to six convolution layers[10]. The local compressing of sparse matrix multiplication and computational power of characteristics significantly increase speed in this system. The fully linked layers are replaced with the average global pooling layer by the Inception module. This accelerates the model and lowers the risk of overfitting [11].

b. AlexNet

In the 2012 ImageNet competition, Krizhevsky et al. introduced the AlexNet architecture, which has 3 fully linked layers and 5 convolutional layers [12]. With 4096 feature vectors and 1000 classes in the photos, AlexNet was created with an input image size of 227x227x3.

c. VGGNet

Architecture of AlexNet is pretty comparable to this. Each block in VGG16 and VGG19 consists of 5. There are two convolution layers and one pooling layer in the first and second blocks, respectively and in blocks 3 and 4, the VGG-16 model uses 3 convolutional layers, while the VGG-19 model uses one pooling layer and four convolutional layers. All convolutional layers are employed with stripe filters of the size 3x3. [13,14].

d. ResNet50

ConvNet (CNN) a kind of neural networks that is most frequently used to analyse visual vision, have a pretrained deep learning model called ResNet-50 for classifying images. ResNet-50, which has 50 layers, was trained with around 1000 classes from the ImageNet database of million images, along with the trainable parameters of around 23-24 millions.

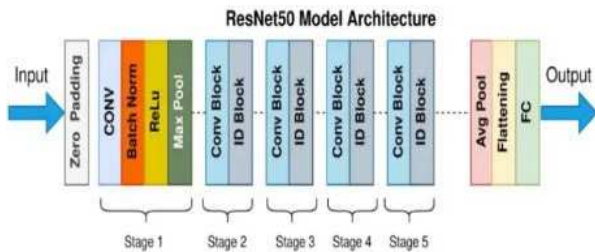


Fig.3 ResNet50 Model Architecture[33]

There are alternative pretrained deep models to utilise, such as AlexNet, GoogleNet, or VGG19, but the ResNet-50 is renowned for having lower fault rates and outstanding classification ability, making it a handy tool to be aware of [16].

This architecture(fig. 3) differs from traditional CNN models like AlexNet and VGGNets since it has 49 convolutional layers and one fully connected layer. The depth of these model's residual blocks and architecture represents its creativity [33]. The input image is 224x224x3 pixels in size. Five 1x1 and 3x3 filters make up the convolutional layer of the algorithm.

The max pooling layer comes after the blocks. Identity mapping is a technique used by residual neural networks, also referred to as ResNets. This means that the input to one layer is given to another directly or as a shortcut. The skip connection or shortcut is the most crucial idea in play here. In essence, a skip connection is an identity mapping as shown in fig. 4. In this the input of the previous layer is added with the output of the next layer. When is input and $F(X)$ is the layer's output, the residual block's output can be given by:

$$y = f(X) + X \quad (3)$$

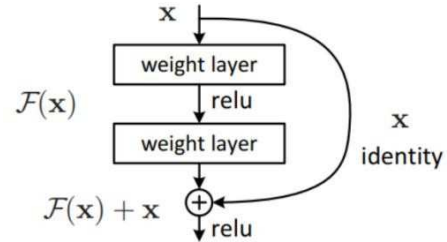


Fig. 4 Basic residual block of ResNet depicting skip connection[34]

A residual deep learning network called ResNet-50 (which has 50 layers) aims to solve the issue of vanishing gradients that happen during the back-propagation of convolutional neural networks (CNN). As long as over-fitting is taken into account, increasing the network's depth should boost its accuracy. However, the issue with increased depth is that because of the increased depth of the network, the signal needed to modify the weights, which is obtained from the end of the network by comparing ground-truth and prediction (observed against forecasted), becomes very little at the earlier layers [2]. As long as over-fitting is taken into account, increasing the network's depth should boost its accuracy. However, the issue with increased depth is that because of the increased depth of the network, the signal needed to modify the weights, which is obtained from the end of the network by comparing ground-truth and prediction (observed against forecasted), becomes very little at the earlier layers. In essence, it indicates that prior levels are still mostly unlearned. When the gradient in the nonlinear optimization that aims to adjust weights approaches close to zero, it is known as the "vanishing gradient" problem. The main factor with training deeper networks is that it adds layers by conducting optimization on a vast parameter space. This increases training error. Such deep networks can be trained using residual networks, which build the network from

known as residual models. This issue is referred to as deterioration.

IV. EXPERIMENTS AND RESULT

Using Keras, a CNN model was created. ANN has an interface for Python thanks to the open-source software framework known as Keras. For the TensorFlow library, Keras serves as an interface. In addition to TensorFlow, Microsoft Cognitive Toolkit, Theano, and PlaidM. On mobile devices, or the Java Virtual Machine, the web, Keras users can productize deep models. Additionally, it permits the deployment of distributed deep learning model training on batch of GPU and (TPU). A high-level API wrapper for Tensorflow is called Keras.

Transfer learning is the process of using the information obtained by finishing one assignment to assist in the solution of a separate but connected challenge. It uses the factors identified when one problem was being solved to find the solution to a somewhat related problem. It applies a previously trained model to a fresh issue. It is typically used for tasks where your dataset contains insufficient data to fully train a model from scratch. Fig. 5 here represents the steps of transfer learning in deep learning, and fig. 6 shows the architecture of ResNet50.

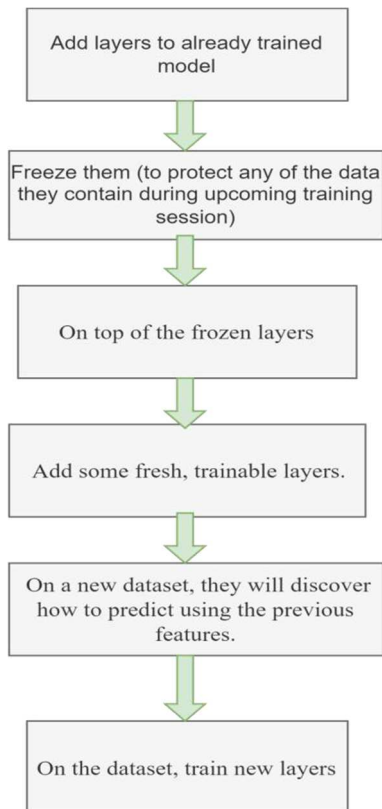


Fig. 5 Flowchart representing transfer learning in deep learning

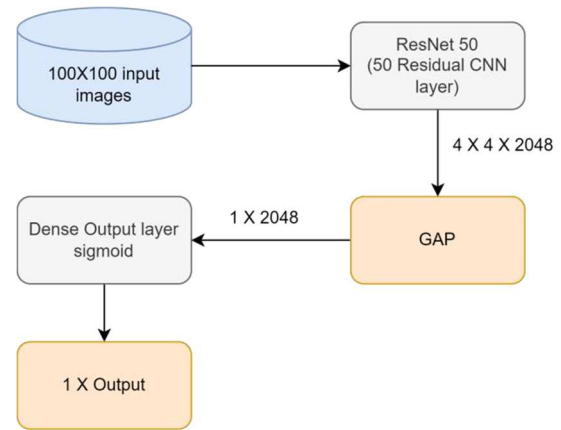


Fig. 6 ResNet50 Architecture (for transfer learning)

B. Proposed Work

- Data collection:** The tympanic membrane images and medical records of patients who visited the otolaryngology section for ear issues was acquired in JPEG format with a suitable resolution of. A correct diagnosis of normal, glue ear, AOM, CSOM, and otomycosis was annotated on each image of the tympanic membrane by three otolaryngologists based on endoscopic images and corresponding medical records, such as operation records or audiologic.
- Preprocessing and augmentation:** Unnecessary black margins are present on the boundaries of unrefined endoscope-captured pictures of the tympanic membrane, necessitating the use of an appropriate preprocessing procedure to eliminate these regions and speed up inference. For data preprocessing, we therefore first reduced the original image's black edges to just include the tympanic membrane before resizing it to a required dimension.
- Data Splitting:** The data set was initially split into 50% each for train and test and then into training and test groups of 70% each, with the results being evaluated individually.
- Tuning Process Using ResNet50:** For ResNet50, images were scaled to 227x277x3. The learning rate for CNN structures was set at a suitable parameter. For optimisation the stochastic gradient descent approach was employed and the mini-batch size equals 4. Trial and error was used to choose the hyperparameters.
- Classification:** After all those steps, the model was able to classify the images into 6 classes AOM, CSOM, glue ear, otomycosis, normal ear and others. The model identifies the disease from those classes. Flow chart representing the steps of proposed methodology is given shown in fig. 7.

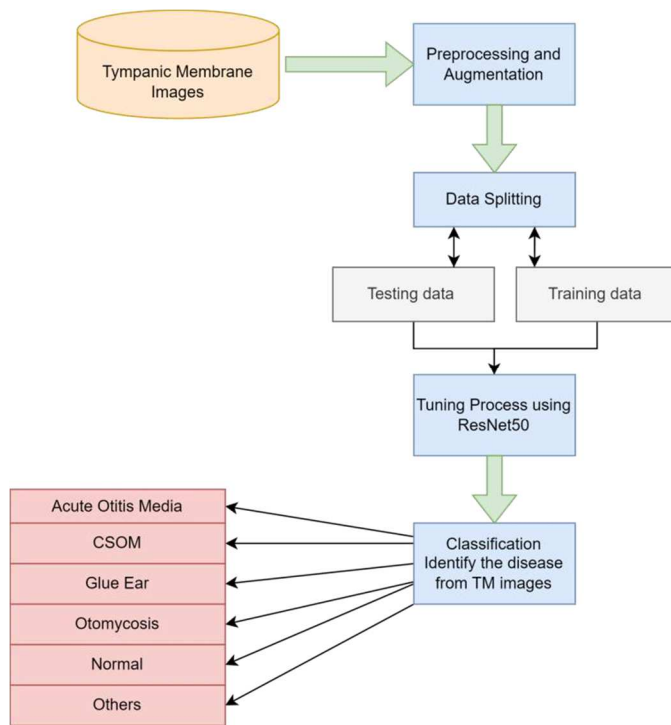


Fig. 7 Flowchart depicting the proposed work

For ResNet50, images were scaled to 227x277x3. Trial and error was used to choose the hyperparameters. The data set was initially split into 50% each for train and test and then into training and test groups of 70% each, with the results being evaluated individually.

B. Results

The data set was splitted into two groups: 50% for training-testing each, or in the ratio of 70:30 for training and testing. The layers of the CNN architecture were then used to analyze the performance data. First, the conv5 layer of the AlexNet architecture obtained 33% and 42% accuracy, respectively. The VGGNets was developed using VGG-19, another architecture. With accuracy rates of 49% and 44% , the conv5 1 produced the best results in this design. Inception 5a-output layer accuracy rates in the GoogLeNet architecture were between 57% and 64%. Using the ResNet50 architecture accuracy results with the activation 49 relu layer of 82% and 84% were achieved. With the increase in the amount of data in second experimental study, the hit rate was incremented in general, but it was observed that it was diminished in a few layers of ResNet50 models and VGG16 models. The ConvNet capacity for learning has increased along with the volume of data. This circumstance backs up the research in the literature [17]. The empirical work used a data set that included 148 abnormal TMs and 90 abnormal TMs. The otoscope gives the images in these classes a variety of hues, of diseased region biomedical images, the deep learning-based methods are discovered to be exceptional when these studies are examined

[35]. The data in the data set are chosen at random in order to evaluate the proposed model's accuracy. ResNet50 activation 49 Relu layer was able to detect the membrane region with an accuracy rate of 84%, by splitting the dataset in the ratio of 70:30 for training and testing.

Figure number 8(a) shows the model accuracy when the data was splitted 50% for training-testing each, and figure no 8(b) shows the model loss for the same.

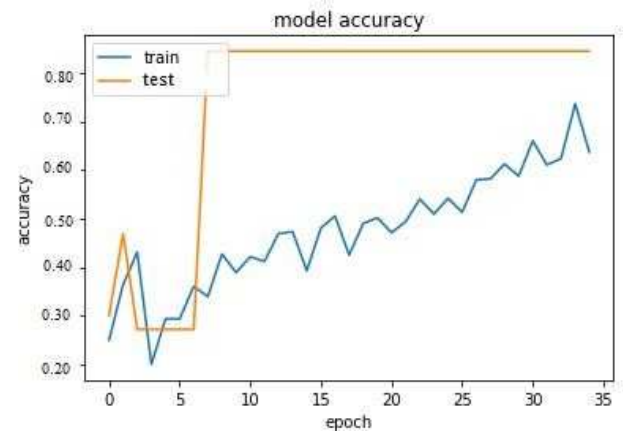


Fig. 8(a) Graph depicting Model Accuracy

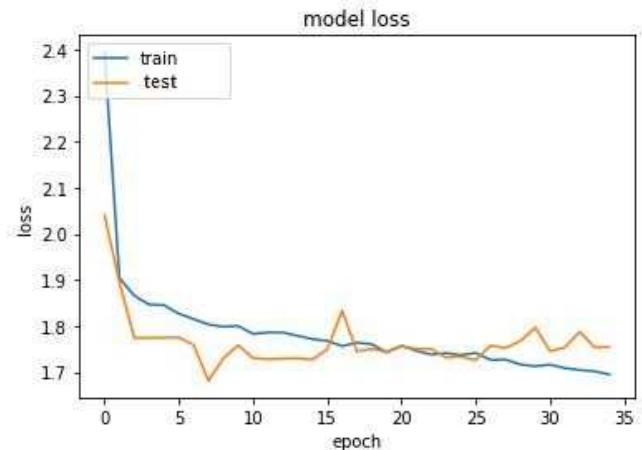


Fig. 8(b) Graph depicting model loss

Fig 9 (a) shows the accuracy of ResNet50 with other CNN Models when the dataset was splitted into the ratio of 70:30 for test and train, and model loss for the same is shown in fig 9(b).

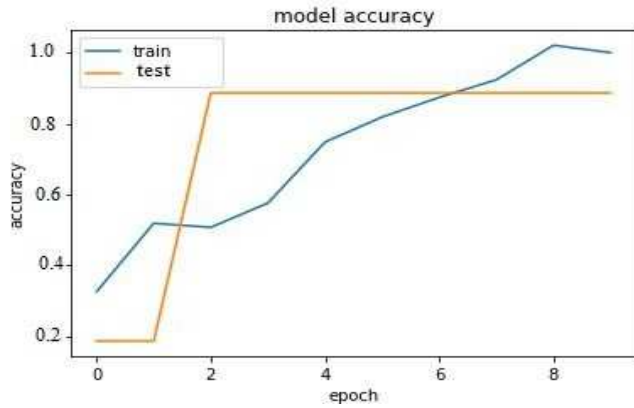


Fig. 9(a) Graph depicting model accuracy

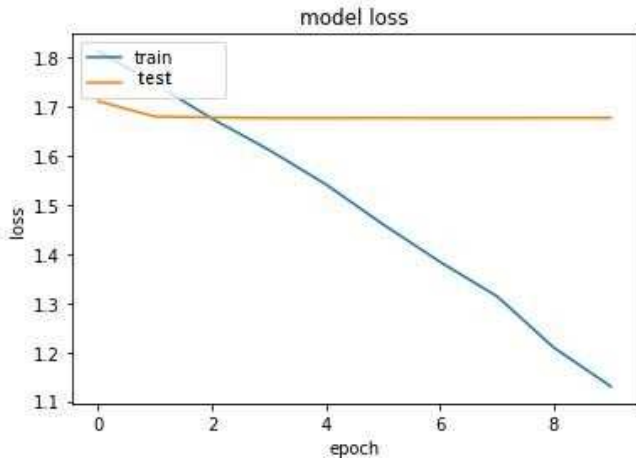


Fig. 9(b) Graph depicting Model loss

TABLE-II

Architecture	Layer	Accuracy
AlexNet	Conv5	33
VGG19	Conv5_1	49
GoogLeNet	Inception_5a	57
ResNet50	Activation49_RelU	82

Table-II and the bar graph in fig. 10(a) shows the comparisons of the accuracy of ResNet50 with other CNN Models when the dataset was divided into 50:50 ratio for training and testing each.

Table-III and the bar graph in fig. 10(b) shows the comparisons of the accuracy of ResNet50 with other CNN Models when the dataset was splitted into the ratio of 70:30 for test and train.

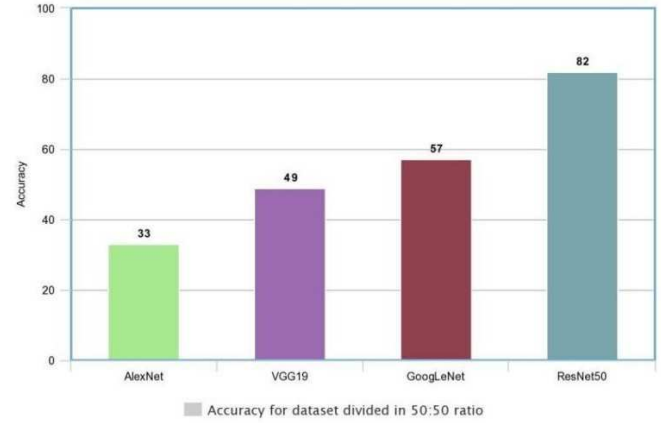


Fig. 10(a) Bar graph showing accuracy for 50:50

TABLE-III

Architecture	Layer	Accuracy
AlexNet	Conv3	42
VGG19	Conv5_1	44
GoogLeNet	Inception_5a	64
ResNet50	Activation49_RelU	84

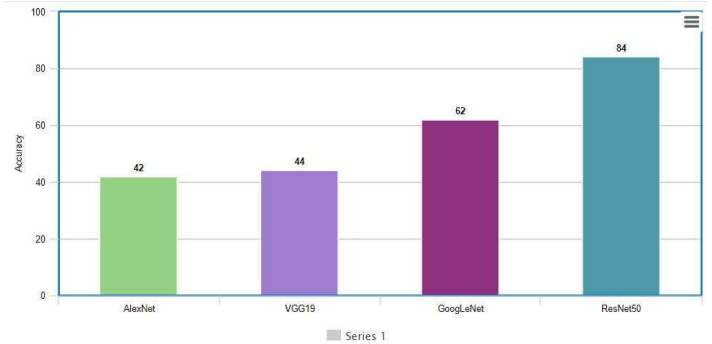


Fig. 10(b) Bar graph showing accuracy for 70:30

V. CONCLUSION AND FUTURE WORK

A two-stage CNN technique for classifying otoscopic images was suggested. Attention mechanism is used by the deep neural learning network to concentrate on the TM conspicuous lesion sites while correctly aligning the images to lessen the effect of noise. The findings demonstrate the classification ability to recognise normal, OME, CSOM, AOM, Otomycosis from otoscope images at the diagnostic level.

Approaches like k-Means are found to be the traditional methods for object detection when looking into the literature. Although classical and deep neural network detection methods are successful for object and region detection like face detection and detection brightness, and darkness. The success rate is affected if the data in the dataset is not distributed evenly. [18]. After a thorough inspection of the temporal lobe, the patient's ear must be diagnosed and treated in order to determine the presence of OM. In experimental studies, 238 pictures were utilised to identify the membrane region in images of the middle ear. To identify the tympanic membrane (eardrum) in these images taken via an auriscope device, a deep learning-based method was employed.

Future research will make use of a data set that includes both normal and AOM TM middle ear pictures, where the membrane region may be observed.

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